



Developing Adaptive Business Intelligence Systems for Agile Supply Chain Analytics

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ABSTRACT

Agile supply chain analytics is about ability to adapt business intelligence (BI) systems to meet the changing needs of the supply network and its environment. Agile BI incorporates both technical and process-oriented approach to development, management, and delivery of BI solutions. To achieve agile BI, supply chains must provide and establish the necessary infrastructure, architecture, tools and processes in a way that enables them to deliver BI systems capable of quickly and effectively adapting to the rapidly changing business needs and environment factors. However, most supply chains still lack methods, processes and tools to successfully design and implement these systems. The paper first provides an analysis of current situation and a concise background research, followed by the introduction of the integrated and adaptive supply chain business intelligence model. This model enables design of pervasive analytical systems for collaborative decision-making, monitoring and management of the supply network. The main components of the model are described, as well as the architecture and features of the specialized supply chain intelligence web portal which demonstrates usefulness and applicability of the proposed model. This system allows creation of a new breed of flexible and agile BI systems which ultimately results in more efficient, responsive and adaptive supply chains.

CCS Concepts

• **Information systems**→**Information Retrieval**→**Retrieval tasks and goals**→**Business intelligence** • **Information systems**→**Information systems applications**→**Decision support systems**→**Data analytics** • **Applied computing**→**Enterprise computing**→**Business process management**→**Business intelligence** • **Applied computing**→**Operations Research**→**Industry and manufacturing**→**Supply chain management**.

Keywords

Business intelligence; supply chain management; data analytics; enterprise modeling; web portal.

1. INTRODUCTION

The new approach to supply chain management (SCM) means

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that companies must find a way to improve communication and information flow, thereby converting the traditional supply chain into an adaptive and real-time supply network. The theory is that this will allow companies to realize the holy grail of the supply chain - a holistic, responsive and flexible management of a network of supply chain resources that improve production and increase profitability [1].

During the transition to adaptive networks, companies must map the three key process enablers – the management of visibility, velocity, and variability – to the three key information enablers – quality, timeliness, and depth of information.

This means that supply networks need to be not only cost-effective, but also to be [2]:

- **Agile** – Respond quickly to disruptions and unexpected changes in business environment and within the supply network. They need to be able to alter processes on demand and meet short term necessities better than other networks.
- **Aligned** – Interest of all supply network partners need to be aligned with the global supply network strategy. Global approach and collaborative planning and decision making are the keys to successful coordination.
- **Adaptable** – Supply network need to evolve over time and to adapt processes to meet other partner, key customer, and changing market needs.

Today, there are variety of business initiatives and technologies such as joint planning and execution, business intelligence, performance management, data mining and alerting that can be used for more efficient supply chain management. However, organizations still lack methods, processes and tools to successfully design and implement these systems.

Business intelligence is an umbrella term that encompasses the processes, tools, and technologies required to turn data into information, and information into knowledge and plans that drive effective business activity [3]. BI encompasses data warehousing technologies and processes on the back end, and query, reporting, analysis, and information delivery tools and processes on the front end.

In order to achieve required supply chain agility and adaptivity, it is necessary to use intelligent technologies and tools which enable monitoring and evaluation of supply chain performance. Combination of business intelligence and performance management systems can improve supply chain efficiency and accountability, and reduce costs with optimized decision-making process based on monitoring of the key performance indicators. In

addition, these systems should enable more predictable performance management by providing actionable information to the right decision makers.

Agile supply chain business intelligence is about an organization's ability to adapt its BI program to meet the changing needs of the s and its operating environment. Agile BI incorporates both a technical and process-oriented approach to managing the delivery of BI projects.

To achieve agile BI, organizations must provide and establish the necessary tools and conditions in a manner that enables them to deliver a BI project capable of quickly and effectively responding to the changing and imprecisely articulated needs of the BI user [4].

In the subsequent sections, we provide background information on supply chain intelligence, and introduce the complete supply chain intelligence model that unifies different elements such as process model, metrics, BI tools and web technologies into the integrated SCI system.

2. BACKGROUND RESEARCH

Given the increasing competition in today's tough business climate, it is vital that organizations provide cost-effective and rapid access to business information for a wide range of business users, if these organizations are to survive into the new millennium. The solution to this issue is a business intelligence system, which provides a set of technologies and products for supplying users with the information they need to answer business questions, and make tactical and strategic supply chain decisions.

In order to be able to read and respond to events in the right time frame with an optimal set of actions requires the ability to be able to be adaptive in processes, collaborative in relationships and above all to be able to use business intelligence to determine when and what to react to.

Investment in business intelligence software is necessary if the organizations want to manage their supply chain more effectively. The studies show that main drivers for BI initiatives include complexity of operations, huge amounts of data, requirements for cost reduction, increased revenue and efficiency improvement, as well as high exposure to risks [5]. Data warehousing and online analytical processing technologies combined with tools for data mining and knowledge discovery have allowed the creation of systems to support organizational decision making [6].

BI systems have evolved significantly during the last several years, however, majority of companies still possess legacy BI solution which have latency problems and are difficult to maintain. BI solutions were usually developed based on limited set of business requirements and against momentary situation, and the technology was selected without common strategy [7]. This resulted in poorly installed and inflexible propriety BI systems tied to a specific platform. Business value was lesser and the maintenance costs were high.

Traditional BI solutions have a poor reputation for being costly, with slow and inflexible development cycles. Traditional BI tools were designed for a select group of highly technical users, and are incapable of providing widespread user deployment because of their cost and complexity.

Despite BI popularity, companies did not implement these systems systematically and consistently. Very often, different

departments or divisions developed their own data warehouses or data marts, purchased specific BI tools, designed end-user BI applications and defined process metrics [8]. Therefore, even though the level of BI usage generally was increasing, BI solution stayed isolated and disconnected.

Business intelligence features that have an interactive feel and intuitive, browser-based navigation will help spread BI to a wider and deeper range of business end users. This model, called pervasive BI, is defined by accessibility of reporting and analysis tools to non-technical end users across organizations [9]. Although mashups and dashboards are not the only features conducive to pervasive BI, they are a great example of how by integrating familiar Web 2.0 capabilities into a BI solution help more end users have access to the data and information they need. To date, the promises of pervasive business intelligence and web-portal communications with suppliers and customers have been largely unrealized.

In the following sections, we present a comprehensive supply chain intelligence (SCI) model and its realization through a specialized SCI web portal which acts as a unique gateway to different information sources and provide services for supply network coordination, planning, analysis, monitoring and data mining [10]. The model and the supporting software solution are developed having in mind most of the previously identified and described drawbacks of existing systems and characteristics of the future SCI systems.

3. ENTERPRISE MODEL FOR AGILE BI

the core elements of the SCI model are process model with metrics, data warehouse and SCI web portal. The process modeling method and the data warehouse dimensional modeling approach, as well as the layered service-oriented composite web application introduce a new breed of robust and flexible SCI systems that can be relatively easily implemented, customized and personalized. This way, previously separate models, tools and technologies are seamlessly integrated into the complete SCI system.

In order to fully realize these SCI capabilities, we need a supply network-wide business excellence model that will provide consistent framework for establishing, modelling, managing, measuring, and improving supply network processes. Figure 1 illustrates the overall supply chain intelligence model.

The proposed SCI model unifies processes, methodologies and tools into a single business solution. SCI model has been developed in such a way to seamlessly integrate within overall BI and collaboration framework [10].

The SCI model is process-centric, metrics-based and modular. It introduces the new supply network and data warehouse modeling approaches, as well as layered application architecture which enables creation of composite and adaptive BI systems.

In contrast to most other BI models that usually approach supply chain from the functional perspective or treat only internal processes, our approach uses the standardized supply chain process model and metrics. The novel process modeling method enables modeling of any supply network regardless of the number of companies, process types and relationships.

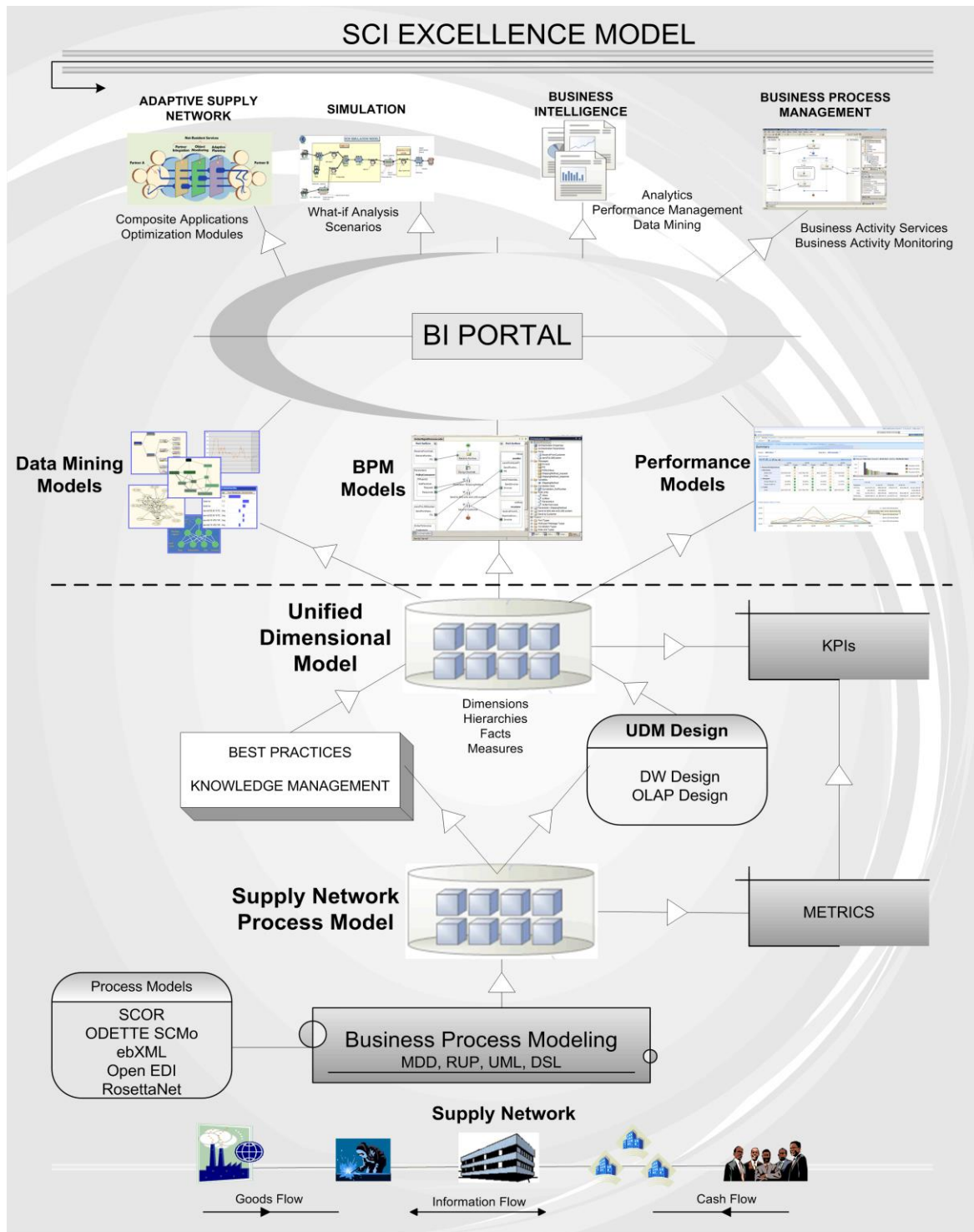


Figure 1. Supply chain excellence model.

Process, metrics, interactions, flows and other elements (for example, resources and constraints) can be modeled at different levels of detail, thus serving as a basis for dimensional modeling, KPI (Key Performance Indicator) design, simulation and business process management.

In order to overcome existing problems with traditional BI systems, we constructed the Unified Dimensional Model (UDM) [11]. The role of a UDM is to provide a bridge between the user

and the data sources. A UDM is constructed on top of one or more physical data sources. It is possible to issue queries against the UDM using a variety of client tools.

Construction of the UDM as an additional layer over the data sources offers a clearer data model, isolation from heterogeneous data platforms and formats, and improved performance for aggregated queries. UDM also allows business rules to be embedded in the model, as well as the option to define actions in

relation to query results. The unified dimensional model serves as an extra layer that isolates and logically unites heterogeneous data sources (such as relational databases, data marts, spreadsheets, web services, etc.) into the single data resource. Also, comparing to traditional DW approaches, the unified dimensional modeling approach allows design of additional DW elements such as complex calculation, hierarchies, measure groups, and KPIs.

The data warehouse approach allows the single version of the truth because of the integrated and consolidated data repository. Server-based definition of KPIs offers centralized metadata management and enables creation of various front-end BI and analytical applications. Data warehouse acts as an integrated data source upon which different BI elements are developed such as planning models, performance monitoring systems, reporting and data mining models.

4. ADAPTIVE SCI WEB PORTAL

The main problem with traditional BI tools is problem with delivering information to end-users or other systems. They usually require special client tools, support only certain data sources (for example ERP systems) and they are difficult or impossible to customize or integrate. That is why our SCI model introduces specialized web portals for analytics, monitoring, planning, data mining and simulation. In order to fulfill these functions, BI portal is built as a layered service-oriented application which utilizes the newest web technologies for the special BI tasks.

The concept is based on service-oriented architecture, models and templates, which should enable creation of the new breed of SCI solutions that connects business processes, data, applications, and users into a single integrated system.

The idea is to define BI service-oriented architecture which provides a standard and flexible approach for integration of business processes, analytics, information, business transactions and collaboration tools. This approach enables separation of business processes and applications, and creation of data, application, and collaboration services on demand and based on events. These services can be further arranged into composite BI applications for specific business processes and activities.

In order to realize the proposed SCI model and to benefit from the composition approach, we need to view software systems granularly and per different layers of the architecture. For example, web service can be application layer asset, OLAP cube can be data layer asset, and the dashboard page can be an element of the presentation layer.

We need architecture and a platform which supports application composition and enables deployment of different combinations of BI elements. Application integration and information exchange can be realized through the SOA (Service-Oriented Architecture) technologies.

By its nature, SCI applications and supply network integration are most often performed after the design and implementation of software systems. Structured business processes and traditional business applications cannot provide support for the complex interactions among supply network partners and their software systems. Because of that, we have added an additional layer to the model, and it is the collaboration layer. The architecture of the layered SCI composite application is presented in Figure 2.

This architecture has advantages over classic SOA concept, as SOA typically provides flexibility only in the application layer. It

offers flexibility in all four layers through the composition of software elements.

BI portal is particularly designed having in mind analytical and decision making needs, not only of a single company, but also of the supply network as well.

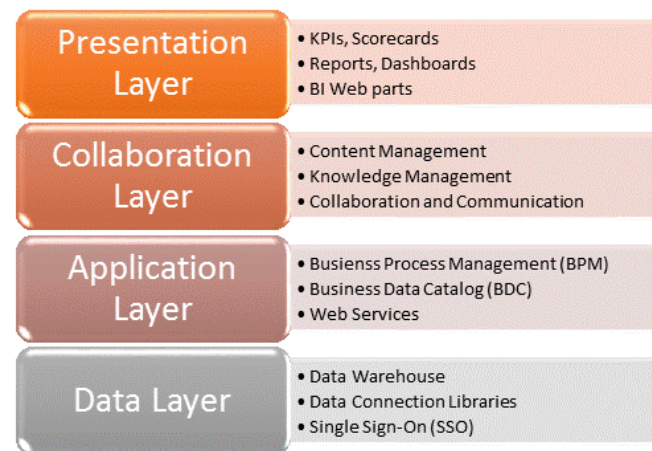


Figure 2. Layers of the SCI composite application.

Portal contains special BI elements which enable information integration and different kinds of analysis.

The main characteristics of the BI web portal are [12]:

- Modularity – ability to compose web pages dynamically integrating content from different sources and using different technologies.
- Personalization – the portal can be easily personalized to reflect users' needs and interests.
- Customization – users can tailor the web portal to suit their specific needs.
- Self-service – users can create new modules, site templates, share and reuse proven solution, and get or access information on-demand.
- Easy user adoption – standardized and consistent browser-based interface enables users to access information using the familiar tools.

The portal represents the single point of access to all relevant information in a personalized and secured manner. Its composite and service-oriented architecture enables inclusion of different BI components and tools (KPIs, dashboards, scorecards, reports, etc.). BI elements can be personalized and adjusted, and information can be filtered just by using a web browser. They can be defined within the portal and also embedded from the external sources via web services. This information is presented through different special web parts. The portal itself can be a provider (via web services or RSS) to other applications.

Figure 3 shows the analytical dashboard page with different web parts that display key performance indicators, charts and graphs. Each web part can have different data source such as OLAP KPI, spreadsheets, report server, web portal database data feeds, etc. Since each data source can reside at different supply chain company, there is a specialized data gateway that acts as a bridge, providing quick and secure data transfer and refresh between the analytical service and on-premises data sources. Such dashboards and reports can be embedded in SCI portals along with other BI web parts thus composing flexible and customized analytical solutions.

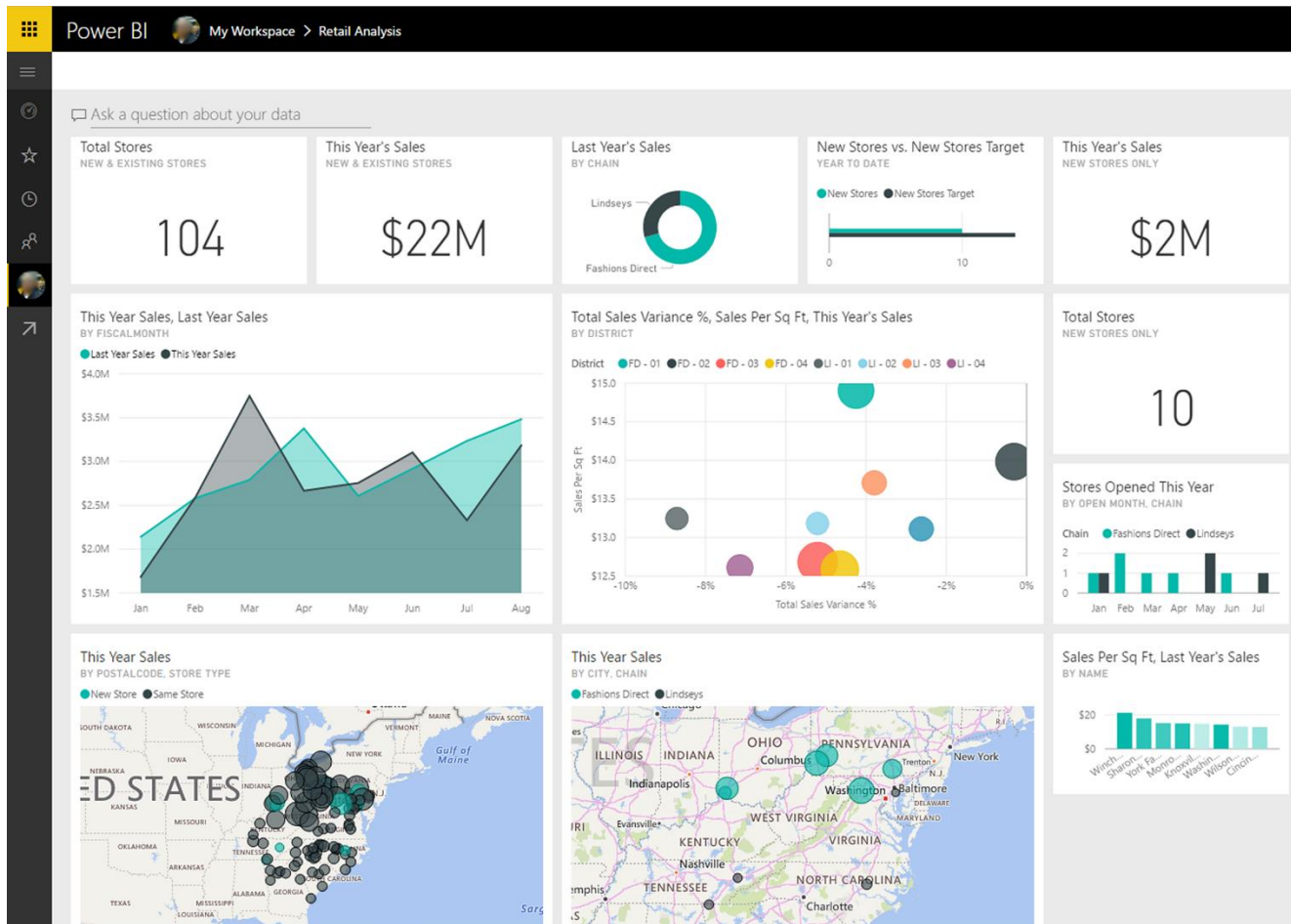


Figure 3. SCI dashboard page.

A system of templates and taxonomies enables creation of many hierarchical portals. For instance, it is possible to create a global supply network BI portal, several company-level BI subportals, and many division-level subportals.

Self-service BI is enabled through the incorporation of special analytical models that enable end-users to create various reports and queries without additional intervention of the IT staff.

All these capabilities make the proposed SCI model and the software system extremely agile, flexible, and applicable in various supply chain scenarios and different industries. Being based on the SCM Metamodel and standardized metrics, ensures general applicability in various business domains.

5. CONSLUSION

The predictable, reliable and efficient movement of materials, components and finished products is critical to the success of every manufacturer, supplier, distributor and retailer. The importance of optimizing this supply network has made supply chain intelligence one of the fastest growing business intelligence and SCM application areas.

In this chapter, the integrated supply chain intelligence model that enables creation of agile analytical systems for collaborative planning, monitoring and management of the supply network is presented.

SCI model fuses all relevant elements such as people, business processes, metrics, KPIs and BI tools into a single comprehensive

system. The underlying architecture supports the complete cycle of BI/PM processes like ETL, DW, OLAP and reporting. The unified data model helps to establish a single version of the truth on supply network performance, which is quantifiable and understood by all entities in the supply network.

SCI web portal is the unifying component that provides integrated analytical information and services, and also fosters collaborative decision making.

The proposed layered SCI architecture allows construction of the next-generation loosely-coupled SCI networks that are built as BI web services. This will enable collaborative, efficient, responsive, and adaptive supply chains.

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